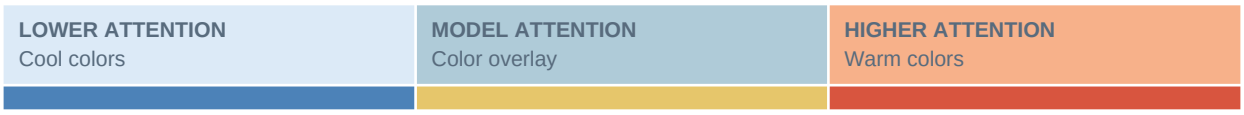
THORAXSCAN

A student guide to attention heatmaps

How to view, interpret, and discuss the model’s visual attention during learning and case review



Key idea: the heatmap is a visual explanation of where the model’s attention was concentrated for a prediction. It is a learning aid. It does not confirm a finding, explain cause, or replace your own image review.

Based on the supplied ThoraxScan student and professor interface frames, including Attention Maps, Learning Mode, activity evaluation, and professor review screens.

1. What the colors mean

ThoraxScan’s Attention Maps screen describes warm colors (red/orange) as higher attention and cool colors (blue) as lower attention. The map overlays a scan to help you notice which image regions received more model attention. Read the color scale shown in the interface from high to low; colors are relative to the visualization and should not be treated as a calibrated probability or a measurement of disease.

On the map	Use it to
Warm / brighter areas	Identify regions the model weighted more strongly in this visualization.
Cooler / less bright areas	Identify regions receiving less displayed attention. This does not prove the area is normal or irrelevant.
Image outside highlighted regions	Keep reviewing it. A heatmap may not show every clinically or educationally relevant feature.

What it does not mean

A bright patch is not automatically a lesion. Attention is not the same as model confidence, diagnostic certainty, or causal evidence. The heatmap shows a pattern associated with the model’s processing; it cannot by itself tell you why the model predicted a class or whether the prediction is correct.

2. Student workflow

Standalone Attention Maps (Figure 7.0)

1. Open **Attention Maps** from the main menu.
2. Select a scan or sample case in the Attention Maps view.
3. Inspect the original X-ray and the attention overlay together. Use the high-to-low legend shown on screen.
4. Describe the location carefully (for example, “attention is concentrated near the lower right lung field”) without naming a disease based on color alone.
5. Compare what you see with the image and the educational explanation, and note uncertainty or areas that do not match your observation.

Learning Mode (Figure 6.0)

Choose a sample or upload a practice image, attempt your own observation first, then reveal the AI attention heatmap explanation for guided feedback. Use the heatmap to compare your visual search with the model’s displayed focus. Learning Mode is described in the frame as ungraded practice.

3. Activity and professor review

Student activity flow (Figures 5.0, 5.2, 5.3)

In an assigned case, review the X-ray and enter your own evaluation before using the attention-map feedback step. When available, select **View Attention Map Feedback** and compare the highlighted region with your written observations. Then consider the professor's evaluation, grade, and comments. The map should support reflection, not replace the assignment's evidence-based answer.

Professor flow (Figures 11.0-11.3.1 and 12.0-13.0)

The professor screens show a sequence for uploading a chest X-ray, running AI inference, viewing the attention map, recording an evaluation, configuring/publishing an activity, and reviewing learning or student feedback. In teaching, ask students to make and record their own image observations first; then use the map as a prompt to discuss agreement, disagreement, and limitations.

Useful discussion prompts

- What anatomy or region is highlighted?
- Does the highlighted area correspond to a visible feature in the X-ray?
- What did you observe outside the highlighted area?
- Where does your interpretation differ from the model's focus, and what should you re-check?

4. A reliable interpretation routine

Observe → **locate** → **compare** → **qualify.** First describe what is visible on the X-ray. Next state where the overlay is strongest. Compare the model's focus with your image-based observation. Finally qualify the conclusion: note agreement or mismatch, and avoid treating attention alone as diagnostic evidence.

Do	Avoid
Use the heatmap as one visual cue alongside the original scan and course knowledge.	Calling a warm spot a confirmed abnormality.
Use location words and describe uncertainty.	Assuming low attention means a region is disease-free.
Discuss mismatches as a chance to re-check the image and model output.	Treating attention as the model's reasoning or proof of causation.

5. Important use limits

ThoraxScan's interface labels the system an assistive tool and says it is not a standalone diagnosis. The Learning Mode frame states that the tool is for Radiologic Technology student learning and that AI output must not be used for clinical diagnosis or patient care. Follow the instructor's directions and use course-approved image interpretation methods.

Quick reference

Warm colors: higher displayed attention. **Cool colors:** lower displayed attention. **Best use:** compare the model's focus with your own image review. **Remember:** the heatmap is an educational visualization, not a diagnosis.